

Further Mechanics 1

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Chapter 4: Elastic Collisions in One Dimension

Why this chapter matters

Why this matters. In Chapter 1, conservation of momentum gave one equation for a collision — not enough to find two unknown final velocities. Newton's law of restitution supplies the missing second equation, unlocking a whole family of exam questions: single collisions, collisions with a wall, and multi-stage “successive collisions” problems that are a Further Maths signature.

Key vocabulary

Term	Meaning
Coefficient of restitution (e)	A number between 0 and 1 (inclusive) describing how “bouncy” a collision is; $e = 1$ is perfectly elastic (no kinetic energy lost), $e = 0$ is perfectly inelastic (particles coalesce, maximum KE lost)
Newton's law of restitution (NLR)	The speed of separation of two colliding particles is e times their speed of approach
Line of centres	In 1D collisions this is simply the common line of motion; it becomes essential terminology in Chapter 5
Successive collisions	A sequence of collisions (e.g. A hits B, then B goes on to hit C) analysed one collision at a time, in order

Key formula table

Quantity	Formula & notes
Conservation of momentum	$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$ — as in Chapter 1
Newton's law of restitution	$v_2 - v_1 = e(u_1 - u_2)$ — speed of separation = $e \times$ speed of approach; sign convention must match momentum equation
Range of e	$0 \leq e \leq 1$ — $e = 1$ perfectly elastic, $e = 0$ particles coalesce
Collision with a fixed wall/smooth plane	$v = eu$ (speed reverses direction, scaled by e) — wall treated as infinite mass, stays at rest
KE lost in a collision	$\frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2 - (\frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2)$ — always ≥ 0 unless $e = 1$,

when it equals 0

Concept: Newton's law of restitution

For two particles colliding directly, with speeds u_1, u_2 before and v_1, v_2 after (all measured in the *same* fixed positive direction):

$$v_2 - v_1 = e(u_1 - u_2)$$

This says: (speed of separation after) = $e \times$ (speed of approach before). Combined with conservation of momentum, this gives two simultaneous equations in the two unknowns v_1 and v_2 — always solvable.

Physical intuition. The coefficient e measures how much kinetic energy a collision destroys. If $e = 1$, the particles separate exactly as fast as they approached — a perfectly elastic collision in which no KE is lost. If $e = 0$, they have zero separation speed (they move off together, i.e. coalesce), which turns out to lose the maximum KE consistent with conserving momentum. Values strictly between 0 and 1 — typical of real collisions like a dropped ball bouncing — mean some KE is converted to heat, sound and deformation on impact; the closer e is to 1, the “bouncier” the collision. One important consequence: e belongs to the *pair* of colliding surfaces, not to a single particle — a particle can have a different e when it hits different objects (as in Worked example 3, where A–B and B–C have different values of e).

Worked example 1 — standard direct collision using both equations

EXAMPLE 4.1

Question. A particle A of mass 4 kg moving at 6 m s^{-1} collides directly with a particle B of mass 2 kg moving at 1 m s^{-1} in the same direction. The coefficient of restitution between A and B is 0.5. Find the velocities of A and B after the collision, and determine whether a second collision occurs.

Step 1. Take the common direction of motion as positive: $u_A = 6$, $u_B = 1$. Conservation of momentum:

$$4(6) + 2(1) = 4v_A + 2v_B \quad \text{M1}$$

$$26 = 4v_A + 2v_B \rightarrow 13 = 2v_A + v_B \dots(1) \quad \text{A1}$$

Step 2. Newton's law of restitution, $e = 0.5$:

$$v_B - v_A = 0.5(6 - 1) = 2.5 \dots(2) \quad \text{M1A1}$$

Step 3. Solve simultaneously. From (2), $v_B = v_A + 2.5$. Substitute into (1):

$$13 = 2v_A + v_A + 2.5 \Rightarrow 10.5 = 3v_A \Rightarrow v_A = 3.5 \text{ m s}^{-1} \quad \text{M1A1}$$

$$\text{Then } v_B = 3.5 + 2.5 = 6 \text{ m s}^{-1}. \quad \text{A1}$$

Step 4. Before the collision, A (faster, $u_A = 6$) was behind B (slower, $u_B = 1$), so A was catching up to B. After the collision, $v_B = 6 > v_A = 3.5$, so B is now moving away from A faster than A is moving — B pulls ahead, so no second collision occurs. **A1**

Step 5 (extension — energy check). Total KE before = $\frac{1}{2}(4)(6^2) + \frac{1}{2}(2)(1^2) = 72 + 1 = 73 \text{ J}$. Total KE after = $\frac{1}{2}(4)(3.5^2) + \frac{1}{2}(2)(6^2) = 24.5 + 36 = 60.5 \text{ J}$. **M1**

KE lost = $73 - 60.5 = 12.5 \text{ J}$. **A1** This is positive, as it must be whenever $e < 1$ — some KE is

always dissipated unless the collision is perfectly elastic.

Answer: $v_A = 3.5 \text{ m s}^{-1}$, $v_B = 6 \text{ m s}^{-1}$ (both in the original direction of motion); no second collision occurs since B moves away from A faster than A travels.

EXAM TIP

The “will there be a second collision?” question trips people up because it's a reasoning step, not a calculation. Think of it as: identify which particle is in front (in the direction of travel) after the collision, then check whether the particle behind is faster (catches up → second collision) or slower/equal (no second collision). A simple diagram with two arrows drawn to scale after the collision makes this almost automatic.

Worked example 2 — collision with a fixed wall

EXAMPLE 4.2

Question. A ball of mass 0.3 kg strikes a fixed smooth wall at right angles with speed 8 m s^{-1} . The coefficient of restitution between the ball and the wall is 0.6 . Find the speed of the ball after impact, and the impulse exerted by the wall on the ball.

Step 1. Treat the wall as a particle of “infinite mass” that remains at rest before and after ($u_{\text{wall}} = v_{\text{wall}} = 0$). Newton's law of restitution applied to ball and wall gives, taking the direction towards the wall as positive:

$$\text{speed of separation} = e \times \text{speed of approach} \implies v = e \times 8 = 0.6 \times 8 = 4.8 \text{ m s}^{-1} \quad \text{M1A1}$$

Step 2. The ball rebounds, so after impact it moves at 4.8 m s^{-1} away from the wall (i.e. its velocity is -4.8 in the original positive direction).

Step 3. Impulse on ball = $m(v - u) = 0.3(-4.8 - 8) = 0.3 \times (-12.8) = -3.84 \text{ N s}$. **M1A1**

Magnitude 3.84 N s , direction away from the wall.

Answer: rebound speed 4.8 m s^{-1} ; impulse on the ball from the wall has magnitude 3.84 N s , directed away from the wall.

EXAM TIP

It is tempting to apply conservation of momentum to a ball-and-wall collision, which is confusing because the wall's mass is not given. Remember that a fixed wall is modelled as having effectively infinite mass and zero velocity throughout — momentum of the “wall + ball” system is not conserved in the usual sense (the wall/floor beneath it absorbs momentum), so for a wall collision you use Newton's law of restitution alone, with the wall's speed fixed at zero.

Worked example 3 — successive collisions

EXAMPLE 4.3

Question. Three particles A, B, C, each of mass 1 kg , lie at rest in a straight line on a smooth horizontal table, with B between A and C. Particle A is projected towards B with speed 5 m s^{-1} . The coefficient of restitution between A and B is 0.5 , and between B and C is 0.8 . Find the final

velocities of A, B and C, and determine whether A collides with B again.

Step 1 (collision 1: A and B). $u_A = 5$, $u_B = 0$. Conservation of momentum (masses equal, so they cancel):

$$5 = v_A + v_B \dots (1) \quad \text{M1A1}$$

$$\text{Restitution: } v_B - v_A = 0.5(5 - 0) = 2.5 \dots (2) \quad \text{M1A1}$$

Solve simultaneously. From (1), $v_A = 5 - v_B$. Substitute into (2):

$$v_B - (5 - v_B) = 2.5 \implies 2v_B = 7.5 \implies v_B = 3.75 \text{ m s}^{-1}, \text{ then } v_A = 1.25 \text{ m s}^{-1} \quad \text{A1}$$

Step 2 (collision 2: B and C). B now moves at 3.75 m s^{-1} towards C (at rest). $u_B = 3.75$, $u_C = 0$. Conservation of momentum:

$$3.75 = v_{B'} + v_C \dots (3) \quad \text{M1}$$

$$\text{Restitution: } v_C - v_{B'} = 0.8(3.75 - 0) = 3 \dots (4) \quad \text{M1A1}$$

$$\text{Adding (3) and (4): } 2v_C = 6.75 \implies v_C = 3.375 \text{ m s}^{-1}, \text{ then } v_{B'} = 0.375 \text{ m s}^{-1} \quad \text{A1}$$

Step 3 (check A and B again). After collision 1, A travels at 1.25 m s^{-1} (unaffected by collision 2). After collision 2, B travels at $v_{B'} = 0.375 \text{ m s}^{-1}$, both in the original direction of motion, with A behind B. Since $v_A = 1.25 > v_{B'} = 0.375$, A is faster than B and catches up, so a second A–B collision **does** occur. **A1**

Answer: $v_A = 1.25 \text{ m s}^{-1}$, v_B (after collision with A) $= 3.75 \text{ m s}^{-1}$, then after B–C collision: $v_{B'} = 0.375 \text{ m s}^{-1}$, $v_C = 3.375 \text{ m s}^{-1}$ (all in A's original direction); a second collision between A and B does occur since A ends up faster than B.

EXAM TIP

This worked example is deliberately chosen to show a case where the “no further collision” conclusion is *not* automatic — successive-collision questions are exactly where exam boards test whether you actually compare speeds rather than assuming the story stops after two collisions. End every successive-collision question with an explicit inequality comparing the relevant two speeds (trailing particle's speed vs leading particle's speed, in the same direction), and to state the physical conclusion in words, not just numbers.

Chapter summary

Term	Meaning
Newton's law of restitution	$v_2 - v_1 = e(u_1 - u_2)$; speed of separation = $e \times$ speed of approach
Combined with momentum	two equations, two unknowns (v_1 , v_2); always solve simultaneously
Range of e	$0 \leq e \leq 1$; $e = 1$ elastic (KE conserved), $e = 0$ particles coalesce
Collision with a fixed	wall has zero velocity throughout; use restitution with the wall's speed

wall	fixed at 0
Successive collisions	resolve one collision at a time, in the order they occur, then explicitly test whether a further collision follows

Common exam mistakes

- Writing the restitution equation with velocities in the wrong order or against the wrong sign convention (it must be $v_2 - v_1$, not $v_1 - v_2$, matching the same positive direction used in the momentum equation).
- Trying to apply conservation of momentum to a collision with a fixed wall (the wall is not part of a momentum-conserving system in the usual sense).
- Assuming no further collisions happen without explicitly comparing final speeds and positions.
- Using e values outside $[0, 1]$, or misreading which collision e refers to in a successive-collision problem with different e values at each stage.
- Assuming the same e applies throughout a successive-collision problem — e is a property of the specific pair of objects in contact, so A–B and B–C can (and often do) have different values.
- Forgetting that $e = 1$ corresponds to a perfectly elastic collision (kinetic energy conserved) — useful as a check when a question implies no energy is lost.

You should now be able to

- State and apply Newton's law of restitution, $v_2 - v_1 = e(u_1 - u_2)$.
- Combine restitution with conservation of momentum to find both velocities after a direct collision.
- Analyse a collision between a particle and a fixed wall or smooth plane.
- Work through successive collisions in sequence and correctly determine whether a further collision occurs.